Empirical Methods for Policy Evaluation

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Randomized Experiments and Dynamic Latent Factor Models

1. Randomized control trials (RCTs)
   - ⇒ SUTVA and assignment mechanisms
   - ⇒ Randomization inference
   - ⇒ Randomization designs
   - ⇒ Stratified RCTs and Clustered RCTs
2. Dynamic Latent Factor Models
   - ⇒ Cunha-Nielsen-Williams (2021, ARE)
3. The production of human capital in the early years
   - ⇒ Attanasio-Cattan-Fitzsimons-Meghir-Rubio (AER, 2020)

SUTVA and assignment mechanisms

Causal Inference as a Missing Data Problem

- Population of units, indexed by $i = 1, ..., N$
- Treatment indicator W_i taking values 0 and 1
- For $i \in \{1, ..., N\}$ $\exists$ one realized outcome and one missing potential outcome

$$Y_i^{\text{obs}} = Y_i(W_i) = \begin{cases} Y_i(0) \text{ if } W_i = 0 \\ Y_i(1) \text{ if } W_i = 1 \end{cases}$$

$$Y_i^{\text{miss}} = Y_i(1 - W_i) = \begin{cases} Y_i(1) \text{ if } W_i = 0 \\ Y_i(0) \text{ if } W_i = 1 \end{cases}$$

- Unit-level causal effect $Y_i(1) - Y_i(0)$ is unobserved

The Stable Unit Treatment Value Assumption (SUTVA)

- We will generally need to predict, or impute, the missing potential outcome
- Denote $W_{-i} = (W_j)_{j \neq i}$, SUTVA requires that

$$(Y_i(1), Y_i(0)) \perp W_{-i}$$

1. No interference

   $\Rightarrow$ Potential outcomes for unit i do not vary with the treatment status of any other j

2. Scale Invariance

   $\Rightarrow$ There are no different forms of each treatment, which lead to $\neq$ potential outcomes

Sources of Possible Violations of SUTVA

1. Spillovers and equilibrium effects. E.g.:
   - $\Rightarrow$ Fertilizer in one plot may affect yields in contiguous plots
   - $\Rightarrow$ Immunization efficacy may depend on the number of people immunized
   - $\Rightarrow$ Prob(job) after training may be affected by the number of people trained

2. Hidden variations of treatments. E.g.:
   - $\Rightarrow$ Endogenous compliance to treatment assignment
   - $\Rightarrow$ Unobserved differences in the method of administering the treatment

Assignment Mechanism

- N units, set $\mathbb{W} = \{0,1\}^N$ of N-vectors with all elements equal to 0 or 1

- The assignment mechanism is a function $P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1)) \in [0,1]$ such that

$$\sum_{\mathbf{W}\in\{0,1\}^N} P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1)) = 1$$

$\Rightarrow$ The unit-level assignment probability is

$$p_i(\mathbf{Y}(0), \mathbf{Y}(1)) = \sum_{\mathbf{W}:W_i=1} P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1))$$

$\Rightarrow$ The propensity score is

$$e(x) = \frac{1}{N(x)} \sum_{i:X_i=x} p_i(\mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1))$$

Example of Assignment Mechanism 1

- Two units, and so (2^2) possible values for $\mathbf{W}$

$$\mathbf{W} \in \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$$

$\Rightarrow$ The assignment mechanism is equal to $P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1)) = \frac{1}{4}$

$\Rightarrow$ Unit assignment probability $p_i = p = \frac{1}{2}$ for $i = 1, 2$

$\Rightarrow$ No covariates, and so propensity score $e = p = \frac{1}{2}$

Example of Assignment Mechanism 2

- Two units where only assignments with one treated and one control unit

$\Rightarrow$ The assignment mechanism is

$$P(\mathbf{W}|\mathbf{Y}(0),\mathbf{Y}(1)) = \begin{cases} 1/2 & \text{if } \mathbf{W} \in \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} \\ 0 & \text{if } \mathbf{W} \in \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\} \end{cases}$$

$\Rightarrow$ As before, $e = p = \frac{1}{2}$

Regular Assignment Mechanisms

1. Individualistic

$$p_i(\mathbf{Y}(0), \mathbf{Y}(1)) = q(Y_i(0), Y_i(1)),\ q(\cdot) \in [0, 1]$$

2. Probabilistic

$$0 < p_i(\mathbf{Y}(0), \mathbf{Y}(1)) < 1$$

3. Unconfounded

$$P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1)) = P(\mathbf{W}|\mathbf{Y}'(0), \mathbf{Y}'(1)) = P(\mathbf{W})$$

Randomized Experiments and Observational Studies

- Individualistic + unconfounded implies

$$P(\mathbf{Y}(0), \mathbf{Y}(1)) = c \cdot \prod_{i=1}^{N} q^{W_i}(1-q)^{1-W_i}$$

$\Rightarrow$ The constant c ensures that the probabilities add to unity

$\Rightarrow$ In this case, $p_i = e = q$

- A regular assignment mechanism with known and controlled $q(\cdot)$ is a randomized experiment
- When the functional form of the assignment mechanism is unknown we have an observational study

Randomization Inference

Two Approaches for Inference (Not Just in RCTs)

1. Asymptotic inference (model-based)
   - ⇒ (semi-) Parametric models for the conditional mean of observed outcomes
   - ⇒ Observed outcomes vary through random sampling from a population of units
   - ⇒ Compute the distribution of the test statistic through large-sample properties
2. Randomization inference (design-based)
   - ⇒ Nonparametric; i.e. no restrictions on the distribution of the potential outcomes
   - ⇒ Treatment assignment is the sole source of randomness; observed outcomes are fixed
   - ⇒ The assignment mechanism determines the distribution of the test statistic

Illustrative Example

Unit	Potential Outcomes				
	Cough Frequency (cfa)		Observed Variables		
	$Y_i(0)$	$Y_i(1)$	W_i	X_i (cfp)	Y_i^{obs} (cfa)
1	?	3	1	4	3
2	?	5	1	6	5
3	?	0	1	4	0
4	4	?	0	4	4
5	0	?	0	1	0
6	1	?	0	5	1

The Sharp Null

- The p-value for the sharp null hypothesis that the treatment had no effect on cough frequency is

$$H_0 \; : \; Y_i(0) = Y_i(1) \; \forall i = 1, ..., 6.$$

$\Rightarrow$ The average null $\mathbb{E}[Y_i(0) - Y_i(1)] = 0$ is weaker than the sharp null hypothesis

- Under H_0, all missings in potential outcomes inferred from observed outcomes

- One test statistics from example above is

$$\begin{aligned} T(W, Y^{\text{obs}}) &= \mid \overline{Y}_t^{\text{obs}} - \overline{Y}_c^{\text{obs}} \mid \\ &= \mid (Y_1^{\text{obs}} + Y_2^{\text{obs}} + Y_3^{\text{obs}})/3 - (Y_4^{\text{obs}} + Y_5^{\text{obs}} + Y_6^{\text{obs}})/3 \mid \\ &= \mid 8/3 - 5/3 \mid = 1.00 \end{aligned}$$

Treatment Assignments Generate Test Distribution

- We can re-do this for $\binom{6}{3} = 20$ permutations of treatment assignments
  - $\Rightarrow$ E.g. instead of $W^{\text{obs}} = (1,1,1,0,0,0)$ take $\tilde{W} = (0,1,1,0,0,1)$
- No change in observed outcomes since $Y_i(0) = Y_i(1) = Y_i^{\text{obs}}$ under H_0
- The value of the test statistic for $\tilde{W}$ is

$$\begin{aligned} T(\tilde{W}, Y^{\text{obs}}) &= \mid (Y_2^{\text{obs}} + Y_3^{\text{obs}} + Y_6^{\text{obs}})/3 - (Y_1^{\text{obs}} + Y_4^{\text{obs}} + Y_5^{\text{obs}})/3 \mid \\ &= \mid 6/3 - 7/3 \mid = 0.33 \end{aligned}$$

Exact p-values

- Calculate the value of the statistic for each assignment vector
  - $\Rightarrow$ In previous example, each assignment vector has prior probability=1/20
- How unusual is $T(W, Y^{\text{obs}}) = 1.00$ under the sharp null hypothesis?
- There are 16/20 assignments with $T(\tilde{W}, Y^{\text{obs}}) > T(W, Y^{\text{obs}})$: p-value = 0.80
- In this example, the observed difference could well be due to chance

Approximate p-values

- Recall that the number of distinct values of the treatment vector is $\binom{N_c+N_t}{N_t}$
  - $\Rightarrow$ For instance, if $N = 100$ and $q = 0.5$ then $\dim(\mathbb{W}^+) = e^{29}$
- We thus need to rely on numerical approximations to calculate the p-value
  - $\Rightarrow$ Draw an N-dimensional vector with N_c zeros and N_t ones from $\mathbb{W}^+$
  - $\Rightarrow$ Repeat this process $K-1$ times and approximate the p-value by:

$$\widehat{p} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}_{T^{\text{dif},k} \geq T^{\text{dif,obs}}}$$

  - $\Rightarrow$ With $K > 1,000$ each assignment has a similar probability with and without replacement

Randomization designs

Classic Randomization Designs

- Classic randomization designs can be characterized by restrictions on $\mathbb{W}^+$

Type of Experiment and Design	Number of Possible Assignments Cardinality of $\mathbb{W}^+$	Number of Units (N) in Sample			
		4	8	16	32
Bernoulli trial	2^N	16	256	65,536	4.2×10^9
Completely randomized experiment	$\binom{N}{N/2}$	6	70	12,870	0.6×10^9
Stratified randomized experiment	$\binom{N/2}{N/4}^2$	4	36	4,900	0.2×10^9
Paired randomized experiment	$2^{N/2}$	4	16	256	65,536

Bernoulli Trials (Coin Tossing)

- $p_i = e = q = 0.5, \mathbb{W}^+ = \{0,1\}^N$, dim$(\mathbb{W}^+) = 2^N$

$$P(\mathbf{W}|\mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1)) = 0.5^N$$

- More generally, with $q \neq 0.5$

$$P(\mathbf{W}|\mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1)) = q^{N_t}(1-q)^{N_c}$$

- No way to ensure "enough" treated and control units under each assignment

Completely Randomized Experiments

- Draw N_t units at random, such that $1 \leq N_t \leq N - 1$
- $q = \frac{N_t}{N}$, and $\dim(\mathbb{W}^+) = \binom{N}{N_t}$
- The set of possible assignment vectors is

$$\mathbb{W}^+ = \left\{ \mathbf{W} \in \mathbb{W} \mid \sum_{i=1}^{N} W_i = N_t \right\}$$

- Possible issue with covariate unbalancedness after treatment assignment

Stratified Randomized Experiments

- The population of units is first partitioned into blocks or strata $B_i = B(\boldsymbol{X}_i)$
- Within each block, we conduct a completely randomized experiment
- The set of possible assignment vectors is

$$\mathbb{W}^+ = \left\{ \mathbf{W} \in \mathbb{W} \mid \sum_{i:B_i=j}^{N} W_i = N_t(j) \right\}$$

- More precise inference as units of same type are in different treat arms

Paired Randomized Experiments

- As many units as treatments within each block
- $N(j) = 2$ and $N_t(j) = 1$ for $j = 1, ..., N/2$, so that

$$\mathbb{W}^+ = \left\{ \mathbf{W} \in \mathbb{W} \mid \sum_{i:B_i=j}^{N} W_i = 1 \right\}.$$

- Useful design when N is small and/or J is large

Other Randomization Designs (Popular in Econ)

- Clustered randomized experiments
- Partial population designs
- Randomized saturation designs
- Adaptive randomized experiments

Clustered Randomized Experiments

- Clusters are defined by partitioning the covariate space $G_{ig} = G(\boldsymbol{X}_i)$
- $\overline{W}_g = \sum_{i:G_{ig}=1} \frac{W_i}{N_g} \in \{0,1\}$ is the average value of W_i for units in cluster g
- The assignment mechanisms concerns groups of units (clusters)

$$\mathbb{W}^+ = \left\{ \mathbf{W} \in \mathbb{W} \mid \sum_{g=1}^{G} \overline{W}_g = G_t \right\}$$

- Relax SUTVA within clusters, but maintain it across clusters

Partial Population Designs

- Within-cluster (non-random) program assignment $E_{ig} = \{0, 1\}$
- Write potential outcomes as $Y_i(W_g, E_{ig})$

$$
\begin{aligned}
ATE &= \mathbb{E}(Y_i(1,1) - Y_i(0,1)) = \\
&= \mathbb{E}(Y_i|W_g = 1, E_{ig} = 1) - \mathbb{E}(Y_i|W_g = 0, E_{ig} = 1) \\
ITE &= \mathbb{E}(Y_i(1,0) - Y_i(0,0)) = \\
&= \mathbb{E}(Y_i|W_g = 1, E_{ig} = 0) - \mathbb{E}(Y_i|W_g = 0, E_{ig} = 0)
\end{aligned}
$$

- ATE and ITE depend on $S_g = \sum_{i \in g} E_{i,g}$, which is endogenous

Randomized Saturation Design (Two-Step Randomization)

- Assign each cluster to a treatment saturation, $S_g = \sum_{i \in g} W_{ig} \in [0, 1)$
- Assign each individual to a treatment status $W_{ig} = \{0, 1\}$ according to S_g
- Write potential outcomes as $Y_i(W_{ig}, S_g)$

$$
\begin{aligned}
ATE(s) &= \mathbb{E}(Y_i(1, s) - Y_i(0, 0)) = \\
&\quad + \mathbb{E}(Y_i | W_{ig} = 1, S_g = s) - \mathbb{E}(Y_i | W_{ig} = 1, S_g = 0) \\
ITE(s) &= \mathbb{E}(Y_i(0, s) - Y_i(0, 0)) = \\
&= \mathbb{E}(Y_i | W_{ig} = 0, S_g = s) - \mathbb{E}(Y_i | W_{ig} = 0, S_g = 0)
\end{aligned}
$$

- Optimal combination of clustered and stratified designs

Adaptive Randomized Experiments

- Repeated cross-sections: $(Y_{it}^1, \ldots.., Y_{it}^k)$ are i.i.d. across both i and t
- Treatment $W \in \{1, ..., k\}$, potential outcomes $Y(w)$
- Decide what share p_t^w will be assigned to treatment W in time t
- Posterior beliefs P_t over $Y(w)$ (e.g. thompson sampling):

$$p_t^w = P_t\left(w = \operatorname*{argmax}_{w'} Y(w')\right)$$

Stratified randomized experiments

The Structure of Stratified Randomized Experiments

- Let J be the number of strata/blocks, and $N(j), N_c(j), N_t(j)$
- Let $S_i \in \{1, ..., J\}$ be the stratum for unit i
- Let $B_i(j) = \mathbf{1}_{S_i=j}$ be the stratum indicator for unit i
- The assignment mechanism is

$$P(\mathbf{W}|\mathbf{B}, \mathbf{Y}(0), \mathbf{Y}(1)) = \prod_{j=1}^{J} \binom{N(j)}{N_t(j)}^{-1} \text{ for } \mathbf{W} \in \mathbb{W}^+$$

$\Rightarrow$ $\mathbb{W}^+ = \{\mathbf{W} \in \mathbb{W} \mid \sum_{i=1}^{N} B_i(j) \cdot W_i = N_t(j) \text{ for } j = 1, ..., J\}$

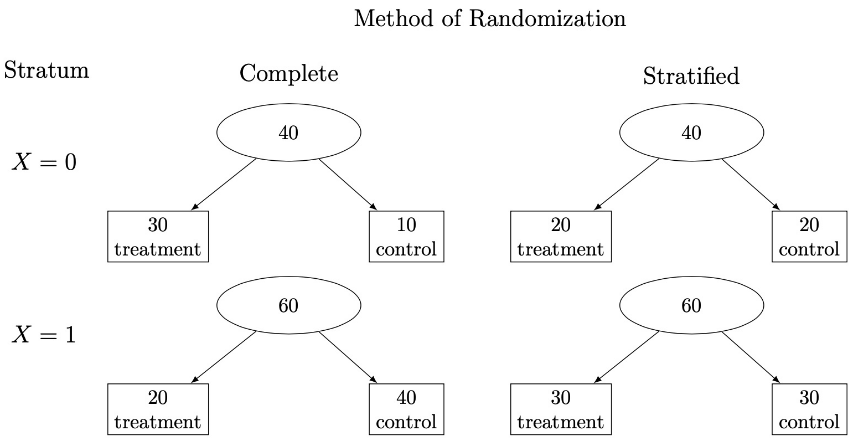
Example: Tennessee Project Star

School/ Stratum	No. of Classes	Regular Classes ($W_i = 0$)	Small Classes ($W_i = 1$)
1	4	−0.197, 0.236	0.165, 0.321
2	4	0.117, 1.190	0.918, −0.202
3	5	−0.496, 0.225	0.341, 0.561, −0.059
4	4	−1.104, −0.956	−0.024, −0.450
5	4	−0.126, 0.106	−0.258, −0.083
6	4	−0.597, −0.495	1.151, 0.707
7	4	0.685, 0.270	0.077, 0.371
8	6	−0.934, −0.633	−0.870, −0.496, −0.444, 0.392
9	4	−0.891, −0.856	−0.568, −1.189
10	4	−0.473, −0.807	−0.727, −0.580
11	4	−0.383, 0.313	−0.533, 0.458
12	5	0.474, 0.140	1.001, 0.102, 0.484
13	4	0.205, 0.296	0.855, 0.509
14	4	0.742, 0.175	0.618, 0.978
15	4	−0.434, −0.293	−0.545, 0.234
16	4	0.355, −0.130	−0.240, −0.150
Average		−0.13	0.09
(S.D.)		(0.56)	(0.61)

The Benefits of Stratification

- DGP with $i = 1, ..., N$, one covariate $X_i \in \{0, 1\}$ and $\pi = \sum_i X_i/N$
- Consider the following two assignment mechanisms for W_i
  - $\Rightarrow$ Complete randomization (CR): $N_t = qN$ and $N_c = (1 - q)N$
  - $\Rightarrow$ Stratified randomization (SBR): $N_{t,X=1} = \pi qN, N_{t,X=0} = (1 - \pi)qN$ and $N_{c,X=1} = \pi(1 - q)N, N_{c,X=0} = (1 - \pi)(1 - q)N$
- SBR reproduces the composition in the overall sample in both $W = 1$ and $W = 0$
- CR generates possible imbalance of W for $X = 1$ and $X = 0$ and higher variance of ATE

Complete Randomization Vs. Stratified: Example with $N = 100$



Randomization Inference for Stratified Experiments

- Sharp null hypothesis

$$H_0 \ : \ Y_i(1) = Y_i(0) \ \forall i = 1, 2, ..., N.$$

- Define average observed outcomes in stratum j as

$$\overline{Y}_t^{\text{obs}}(j) = \frac{1}{N_t(j)} \sum_{i:S_i=j} W_i Y_i^{\text{obs}}$$

$$\overline{Y}_c^{\text{obs}}(j) = \frac{1}{N_c(j)} \sum_{i:S_i=j} (1 - W_i) Y_i^{\text{obs}}$$

- Strata-level propensity score is

$$e(j) = \frac{N_t(j)}{N(j)}$$

Test Statistics

- Within-stratum test statistic

$$T^{\text{dif}}(j) = \left|\overline{Y}_t^{\text{obs}}(j) - \overline{Y}_c^{\text{obs}}(j)\right|$$

$\Rightarrow$ Not very informative as we are interested in treatment effects across all strata

- Linear combination of the within-stratum statistics

$$T^{\text{dif},\lambda_{RSS}} = \left|\sum_{j=1}^{J} \frac{N_j}{N} \left(\overline{Y}_t^{\text{obs}}(j) - \overline{Y}_c^{\text{obs}}(j)\right)\right|$$

$\Rightarrow$ Need $e(j)$ to vary across j for the test to have power over $T^{\text{dif}} = \left|\overline{Y}_t^{\text{obs}} - \overline{Y}_c^{\text{obs}}\right|$

Randomization Inference of the Tennessee Project Star

- $B_i(j), i = 1, ..., 68$ (class-level data)

$$H_0 \ : \ Y_i(1) = Y_i(0) \, \forall i = 1, 2, ..., 68.$$

- Total number of possible assignments is $6^{13} \times 10^2 \times 15$
  - $\Rightarrow$ 13 Schools with two classes in each group: $\binom{4}{2} = 6$
  - $\Rightarrow$ 2 Schools with three small classes and two regular classes: $\binom{5}{2} = 10$
  - $\Rightarrow$ 1 School with four small classes and two regular classes: $\binom{6}{2} = 15$

$\Rightarrow$ $T^{\text{dif}} = \mid \overline{Y}_t^{\text{obs}} - \overline{Y}_c^{\text{obs}} \mid = 0.224$, with $p = 0.034$

$\Rightarrow$ $T^{\text{dif},\lambda_{RSS}} = \mid \sum_{j=1}^{J} \frac{N_j}{N} (\overline{Y}_t^{\text{obs}}(j) - \overline{Y}_c^{\text{obs}}(j)) \mid = 0.241$, with $p = 0.023$

Regression Analysis

$$Y_i^{\text{obs}} = \tau W_i + \sum_{j=1}^{J} \beta(j) B_i(j) + \epsilon_i$$

- $\widehat{\tau}^{\text{ols}}$ is not a consistent estimator of ATE if $\tau(j) \neq \tau(j') \; \forall j \neq j'$

$$WATE = \frac{\sum_{j=1}^{J} \omega(j)\tau(j)}{\sum_{j=1}^{J} \omega(j)}$$

- $\omega(j) = \frac{N_j}{N} \frac{N_t(j)}{N(j)} \frac{N(j) - N_t(j)}{N(j)} = q(j)e(j)(1 - e(j))$

Asymptotic Inference

- The asymptotic variance of the WATE is

$$\mathbb{V}^{\text{strat}} = \frac{\sum_{i=1}^{N} \epsilon_i^2 \cdot \left(W_i - \sum_{j=1}^{J} q(j) B_i(j)\right)^2}{\left(\sum_{j=1}^{J} \omega(j)\right)^2}$$

- Variance weights more those $\widehat{\tau}(j)$ that are more precisely estimated

$$\mathbb{V}^j = \frac{\sigma^2/N}{q(j)e(j)(1-e(j))}$$

Fully-interacted Model

$$Y_i^{\text{obs}} = \tau W_i \frac{B_i(j)}{N(j)/N} + \sum_{j=1}^{J} \beta(j) B_i(j) + \sum_{j=1}^{J-1} \gamma(j) W_i \left(B_i(j) - B_i(J) \frac{N(j)}{N(J)} \right) + \epsilon_i$$

- In this case, $\hat{\tau}^{\text{ols}}$ is a consistent estimator of ATE
- With asymptotic variance

$$\mathbb{V}^{\text{strat,inter}} = \sum_{i=1}^{N} q(j)^2 \cdot \left(\frac{\sigma_c^2(j)}{q(j)(1-e(j))} + \frac{\sigma_t^2(j)}{q(j)e(j)} \right)$$

- In general, $\mathbb{V}^{\text{strat,inter}} > \mathbb{V}^{\text{strat}}$

Regression Analysis of the Tennessee Project Star

- The point estimate of τ in the standard model is
  - $\widehat{\tau}^{\text{ols}} = 0.238$ $(\widehat{s.e.} = 0.103)$
- The point estimate of τ in the fully-interacted model is
  - $\widehat{\tau}^{\text{ols,inter}} = 0.241$ $(\widehat{s.e.} = 0.095)$
- Limited heterogeneity in the treatment effects across strata

Clustered Randomized Experiments

The Structure of Clustered Experiments

- Let G_{ig} be a binary indicator that unit i belongs to cluster $g = 1, ..., G$
- $N_g = \sum_{i=1}^{N} G_{ig}$, so that N_g/N is the share of cluster g in the sample
- $\overline{W}_g \in \{0, 1\}$ is the treatment assignment for all units in cluster g
- The assignment mechanism is

$$P(\mathbf{W}, \mathbf{Y}(0), \mathbf{Y}(1), \mathbf{X}) = \binom{G}{G_t}^{-1}$$

$\Rightarrow$ $\mathbb{W}^+ = \{\mathbf{W} \in \mathbb{W} \mid \sum_{g=1}^{G} \overline{W}_g = G_t\}$

Example: The *Progresa* Program (Again!)

- Clustered RCT during the roll-out of the program in rural areas
  - $\Rightarrow$ 506 villages among those eligible to receive the program
  - $\Rightarrow$ 320 early treatment and 186 late treatment (control)
- Individual/HH level data for both eligible and non-eligible HHs in each village
  - $\Rightarrow$ $\sim$30,000 program eligible children
  - $\Rightarrow$ About 50-100 HHs per village

Estimands

- The choice of estimand depends on the unit of analysis

- Unit-level: a natural estimand is the usual ATE

$$\tau = \frac{1}{N}\sum_{i=1}^{N}(Y_i(1) - Y_i(0))$$

- Cluster-level: weighted average of the unit-level treatment effect

$$\tau^{\text{clust}} = \frac{1}{G}\sum_{g=1}^{G}\tau_g, \quad \text{where } \tau_g = \frac{1}{N_g}\sum_{i:G_{ig}=1}^{N}(Y_i(1) - Y_i(0))$$

Unit-level Vs. Cluster-level

- Cluster-level analysis is more directly linked to the randomization framework
  - $\Rightarrow$ Differently-sized clusters, such as states or towns
  - $\Rightarrow$ Many units will be in the same treatment group so unit-level inference is tricky
- Unit-level incorporates individual-level covariates, which improve efficiency
  - $\Rightarrow$ More homogenous clusters, such as schools or classrooms

Randomization Inference

- The usual statistic for unit-level analysis

$$T^{\text{dif}} = \left| \overline{Y}_t^{\text{obs}} - \overline{Y}_c^{\text{obs}} \right|$$

- The equivalent statistic for cluster-level analysis

$$T^{\text{clust}} = \left| \frac{1}{G_t} \sum_{g:\overline{W}_g=1} \overline{Y}_g^{\text{obs}} - \frac{1}{G_c} \sum_{g:\overline{W}_g=0} \overline{Y}_g^{\text{obs}} \right|$$

Randomization Inference of *Progresa*

- Children-level analysis on school enrollment (pre-program year 1997)

$$T^{\text{dif}} = 0.0075, \quad p\text{-value} = 0.400$$

- Children-level analysis on school enrollment (program year 1998)

$$T^{\text{dif}} = 0.0388, \quad p\text{-value} < 0.001$$

- Village-level analysis on school enrollment (program year 1998)

$$T^{\text{clust}} = 0.0234, \quad p\text{-value} = 0.0120$$

Regression Methods

- In unit-level analysis, we estimate the following regression

$$Y_i^{\text{obs}} = \alpha + \tau W_i + \bar{X}'\gamma + \epsilon_i$$

- $\widehat{\tau}^{\text{ols}}$ consistently estimates ATE (with centered covariates)

- Use Liang-Zeger clustered Var-Cov:

$$\mathbb{V}_{\text{clust}} = \frac{\sum_{g=1}^{G}\left(\sum_{i:G_{ig}=1} \epsilon_i^2 \cdot \left(W_i - \overline{W}\right)^2\right)}{\left(\sum_{i=1}^{N}(W_i - \overline{W})^2\right)^2}$$

Regression Analysis: Cluster-Level

- In cluster-level analysis, consider the following regression

$$\overline{Y}_g^{\text{obs}} = \alpha + \tau \overline{W}_g + \eta_g$$

- $\widehat{\tau}^{\text{ols}}$ consistently estimates ATE

- The sampling variance of τ^{ols} is the usual one

$$\mathbb{V} = \frac{\sum_{g=1}^{G} \eta_g^2}{\sum_{g=1}^{G} \left(\overline{W}_g - \overline{W}\right)^2} = \sigma^2 \left\{ \frac{1}{G_t} + \frac{1}{G_c} \right\}$$

Regression Analysis of *Progresa*

- Children-level analysis on school enrollment (pre-program year 1997)

$$\hat{\tau}^{\text{ols}} = 0.0075 \; (\widehat{s.e.} = 0.0091)$$

- Children-level analysis on school enrollment (program year 1998)

$$\hat{\tau}^{\text{ols}} = 0.0388 \; (\widehat{s.e.} = 0.0104)$$

- Village-level analysis on school enrollment (program year 1998)

$$\hat{\tau}^{\text{ols}} = 0.0234 \; (\widehat{s.e.} = 0.0092)$$

Dynamic Latent Factor Models

The Production Function of Human Capital

$$\boldsymbol{H}_{t+1} = g_t(\boldsymbol{H}_t, \boldsymbol{X}_t, \boldsymbol{I}_{t+1}, u_{t+1})$$

- $\boldsymbol{H}_t = \{\theta_{c,t}, \theta_{s,t}, \theta_{h,t}\}$ is human capital (including cognition, socio-emotional, and health)
- $\boldsymbol{X}_t = \{\theta_{m,t}, \theta_{f,t}, \theta_{r,t}\}$ are parental skills and other background variables
- $\boldsymbol{I}_{t+1} = \{\theta_{M,t}, \theta_{T,t}\}$ are parental investments (including materials and time) between t and $t+1$
- Stocks of current period skills produce next period skills

Self-Productivity and Dynamic Complementarities

- Self-productivity arises when

$$\frac{\partial g_t}{\partial \boldsymbol{H}_t} > 0$$

- Period t stocks of skills promote the acquisition of skills by making investment more productive

$$\frac{\partial^2 g_t}{\partial I_t \partial \boldsymbol{H}_t} > 0$$

- Explains why returns to education are higher at later in child life cycle for more able children

Measurement System

- For each $k \in \{c, n, m, f, ...\}$ and period t we have measures such that:

$$m_{k,t,j} = \alpha_{k,t,j}\theta_{k,t} + \epsilon_{k,t,j}$$

  $\Rightarrow$ $m_{k,t,j}$ is the j^{th} measurement of skill $\theta_{k,t}$

  $\Rightarrow$ $\alpha_{k,t,j}$ is a factor loading

  $\Rightarrow$ $\epsilon_{k,t,j}$ is the measurement error contained in $m_{k,t,j}$

- With two measures per factor, the distributions of factors and measurement error are identified

  $\Rightarrow$ Set the factor scale: $\alpha_{k,t,1} = 1$ + factor location: $E(\theta_{k,t}) = 0, \ \forall t = 1, ...T$

  $\Rightarrow$ Need $J = 2M + 1$ measures to identify factor loadings for M factors

Identification: Factor Loadings

- Assume that the $\epsilon_{k,t,j}$ are uncorrelated across t, then

$$\mathrm{Cov}(m_{k,t,1}, m_{k,t+1,1}) = \mathrm{Cov}(\theta_{k,t}, \theta_{k,t+1})$$
$$\mathrm{Cov}(m_{k,t,2}, m_{k,t+1,1}) = \alpha_{k,t,2}\mathrm{Cov}(\theta_{k,t}, \theta_{k,t+1})$$

- Hence, we can identify the factor loading

$$\frac{\mathrm{Cov}(m_{k,t,1}, m_{k,t+1,1})}{\mathrm{Cov}(m_{k,t,2}, m_{k,t+1,1})} = \alpha_{k,t,2}$$

- if $J > 2M + 1$ one can also allow for serial correlation in the $\epsilon_{k,t,j}$

Identification: Distribution of Latent Variables

- Once $\alpha_{k,t,j}$ are identified, we can re-write the measurement system as

$$\frac{m_{k,t,j}}{\alpha_{k,t,j}} = \theta_{k,t} + \frac{\epsilon_{k,t,j}}{\alpha_{k,t,j}}$$

- The signal is common to multiple measurements, but the noise is not
- To extract the noise from the signal, assume that the $\epsilon_{k,t,j}$:
  1. Are uncorrelated across measures conditional on θ
  2. Are independent from factors

- Then we can identify the joint distribution of $\{\theta_{k,t}\}_{t=1}^{T}$

- Cunha et al (2010) also discuss identification under the more general case with unknown mapping between measures and factors and non-separable measurement errors

The Investment Problem

- Parents choose investments by solving

$$\begin{aligned}\max_{\{C_t, I_{t+1}\}} U(C_t, \boldsymbol{H}_{t+1}) \\ s.t.\ C_t + \boldsymbol{P}_t^I I_{t+1} = Y_t \\ \boldsymbol{H}_{t+1} = g_t(\boldsymbol{H}_t, \boldsymbol{X}_t, \boldsymbol{I}_{t+1}, u_{t+1})\end{aligned}$$

- Investment could be in commodities or in time
- Could easily add parental labor supply decisions
- Could easily add other inter-temporal dimensions beyond HK

Endogeneity of Investment

- Parents choose investment in reaction to shocks, and so $E(u_{t+1}|I_{t+1}) \neq 0$
- Posit a reduced-form of the investment function

$$\boldsymbol{I}_{t+1} = f_t(Y_t, \boldsymbol{P}_t, \boldsymbol{X}_t, e_{t+1})$$

- Notice that Y_t and $\boldsymbol{P}_t$ are excluded from the production function
- One can estimate the investment equation jointly with the production function
- Or use a control function approach to estimate the role of investments in the production function
  - Plug estimated residuals $\hat{e}_{t+1}$ as additional covariates in the production function, so that $E(u_{t+1}|I_{t+1}, \hat{e}_{t+1}) = 0$

Estimation

- Estimation can be performed in two-steps
  1. Estimate the joint distribution of all latent factors
  2. Draws from such a distribution to estimate the investment equation as well as the production function with control function terms
- Inference can be done by using appropriate bootstrapping
  - This takes into account both estimation error at each stage and simulation error

The production of human capital in the early years

Estimating the Production Function for Human Capital: Results from a Randomized Controlled Trial in Colombia

- Human capital formation is a complex process
  - Human capital is multi-dimensional (cognitive, non-cognitive, health,...)
  - Skill formation is a dynamic process
  - The components of human capital interact both within and across periods
  - Both genetic and environmental factors are important inputs

$\Rightarrow$ The early years is a particularly salient period for policy

  - Human capital is malleable (and vulnerable)
  - Dynamic complementarities (skills beget skills)
  - There is no equity-efficiency trade-off for early investment

The Intervention and Its Evaluation

- Adapt a prominent ECD program in Jamaica to the Colombian context
  - Children 12-24 months living in families targeted by a cash transfer program
  - Weekly home visits of one hour from trained women from the community
  - Micronutrients supplementation
  - Cheap and scalable intervention

⇒ Clustered RCT design, stratified by region

- 1,429 children in 96 municipalities in central Colombia
- Municipalities were randomized into 4 groups (cross-cutting or factorial design)

⇒ Control

⇒ Stimulation (home visits)

⇒ Supplementation (micronutrients)

⇒ Stimulation+Supplementation

Rich Data Collected on Children and Parents

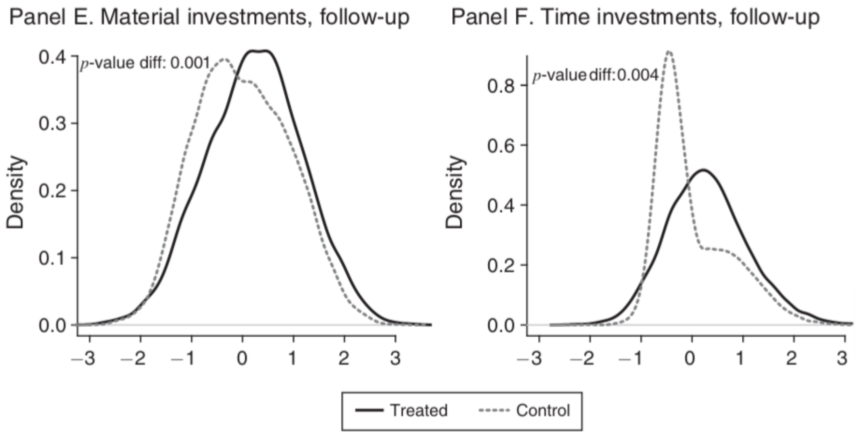
- A battery of psychometrics on children development (baseline and follow-up)
  - ⇒ Motor and cognitive skills: Bayley test
  - ⇒ Socio-emotional skills: Bates temperament test
  - ⇒ Language development: MacArthur-Bates test
  - ⇒ Height, weight, hemoglobin, and morbidity
  - ⇒ Food intakes
  - ⇒ Child care arrangements and time use
- Mothers and families
- Home visitor
- Qualitative instruments

Experimental Results

$\Rightarrow$ Stimulation improved cognitive scores by 0.25σ+receptive language by 0.18σ

- Also positive (noisy) effects on socio-emotional skills
- Micronutrient supplementation had no significant effect on any outcome
- No effect of the intervention on height, weight, or hemoglobin levels

A First Hint at the Mechanisms: Parental Investment



What is the Rationale for the Model?

- Intervention changes human capital through three different channels
  - $\Rightarrow$ Total factor productivity
  - $\Rightarrow$ The productivity of the individual inputs
  - $\Rightarrow$ The distribution of inputs (and in particular parental investment)
- Jointly estimate the distribution of inputs and the production function
- Unpack the role of these different hypotheses

Measurement Equations

$$m^{\theta}_{j,d} = \mu^{k}_{j,d} + \alpha^{k}_{j} \ln \theta^{k}_{d} + \epsilon^{k}_{j}$$

- Treated and control children are allowed to have different factor distributions
- Latent factors are distributed as a mixture of two normal distributions
- Measurement errors terms are jointly normal
- Likelihood function for each group writes as

$$L_i(m) = \int_{\theta} f(m_i|\theta)[\tau \phi_A(\theta) + (1-\tau)\phi_b(\theta)]d\theta$$

$\Rightarrow$ $f(m_i|\theta)$ is the density of the measurement system

Signal-to-noise Ratios: $s_{j,d}^{ln\theta^k} = \frac{(\alpha_j^k)^2 var(ln\theta^k)}{(\alpha_j^k)^2 var(ln\theta^k) + var(\epsilon_j^k)}$

Latent factor	Measurement	Survey	Control	Treatment
Child's cognitive skills at FU (θ_{t+1}^C)	Bayley: Cognitive	FU	0.78	0.79
	Bayley: Receptive language	FU	0.75	0.76
	Bayley: Expressive language	FU	0.78	0.79
	Bayley: Fine motor	FU	0.59	0.61
	MacArthur: Words the child can say	FU	0.64	0.65
	MacArthur: Complex phrases the child can say	FU	0.52	0.54
Child's cognitive skills at BA (θ_t^C)	Bayley: Cognitive	BA	0.70	0.70
	Bayley: Receptive language	BA	0.73	0.72
	Bayley: Expressive language	BA	0.75	0.74
	Bayley: Fine motor	BA	0.60	0.59
	MacArthur: Words the child can say	BA	0.45	0.44
Child's socio-emotional skills at FU (θ_{t+1}^S)	ICQ: Difficult (-)	FU	0.74	0.71
	ICQ: Unsociable (-)	FU	0.33	0.30
	ICQ: Unstoppable (-)	FU	0.59	0.55
	ECBQ: Inhibitory control	FU	0.73	0.69
	ECBQ: Attentional focusing	FU	0.27	0.24
Child's socio-emotional skills at BA (θ_t^S)	ICQ: Difficult (-)	BA	0.68	0.71
	ICQ: Unsociable (-)	BA	0.28	0.31
	ICQ: Unadaptable (-)	BA	0.35	0.38
	ICQ: Unstoppable (-)	BA	0.22	0.25

Treatment Impacts on Measures and Factors

	Treatment effect		
	Point estimate	SE	Sample size
Panel A. Child's cognitive skills at follow-up			
Bayley: Cognitive	0.250	(0.063)	1,264
Bayley: Receptive language	0.175	(0.063)	1,264
Bayley: Expressive language	0.032	(0.062)	1,263
Bayley: Fine motor	0.072	(0.060)	1,262
MacArthur: Words the child can say	0.092	(0.064)	1,322
MacArthur: Complex phrases the child can say	0.058	(0.055)	1,322
Cognitive factor	0.115	(0.051)	
Panel B. Child's socio-emotional skills at follow-up			
ICQ: Difficult (-)	0.074	(0.045)	1,326
ICQ: Unsociable (-)	0.041	(0.054)	1,326
ICQ: Unstoppable (-)	0.032	(0.054)	1,326
ECBQ: Inhibitory control	−0.003	(0.058)	1,323
ECBQ: Attentional focusing	0.069	(0.049)	1,323
Socio-emotional factor	0.087	(0.044)	
Panel C. Material investments at follow-up			
FCI: Number of types of play materials	0.215	(0.064)	1,326
FCI: Number of coloring and drawing books	−0.133	(0.056)	1,326
FCI: Number of toys to learn movement	−0.048	(0.065)	1,326
FCI: Number of toys to learn shapes	0.416	(0.088)	1,326
FCI: Number of shop-bought toys	0.024	(0.061)	1,326
Material investment factor	0.227	(0.069)	
Panel D. Time investments at follow-up			
FCI: Number of types of play activities in last 3 days	0.277	(0.050)	1,326
FCI: Number of times told a story to child in last 3 days	0.138	(0.060)	1,326
FCI: Number of times read to child in last 3 days	0.362	(0.062)	1,326
FCI: Number of times played with toys in last 3 days	0.175	(0.060)	1,326
FCI: Number of times named things to child in last 3 days	0.137	(0.048)	1,326
Time investment factor	0.302	(0.068)	

The Production Function for Human Capital

- Cobb-Douglas specification for the technology of skill formation

$$\ln(\theta^k_{i,d,t+1}) = A^k_d + \gamma^k_{1,d}\ln(\theta^C_{i,t}) + \gamma^k_{2,d}\ln(\theta^S_{i,t}) + \gamma^k_{3,d}\ln(P^C_{i,t}) + \gamma^k_{4,d}\ln(P^S_{i,t}) \\ + \gamma^k_{5,d}\ln(I^M_{i,t+1}) + \gamma^k_{6,d}\ln(I^T_{i,t+1}) + \gamma^k_{7,d}\ln(n_{i,t}) + \eta^k_{i,t+1}$$

- A^k_d captures the direct effect of the home-visitor during her weekly visit

$\Rightarrow$ All parameters are skill-specific and depend on treatment status

Parental Investments

- Same log-linear specification for investment equation ($\tau = \{M, T\}$)

$$\ln(I^{\tau}_{i,t+1}) = \lambda^{\tau}_{0,d} + \lambda^{\tau}_{1,d}\ln(\theta^{C}_{i,t}) + \lambda^{\tau}_{2,d}\ln(\theta^{S}_{i,t}) + \lambda^{\tau}_{3,d}\ln(P^{C}_{i}) + \lambda^{\tau}_{4,d}\ln(P^{S}_{i}) + \lambda^{\tau}_{5,d}\ln(n_{i,t}) + \lambda^{\tau}_{6,d}Z_{i,t} + u_{i,t+1}$$

- Z_t are instrument
- All parameters are allowed to vary with the treatment
- Control function to account for correlation between shocks and investments

$\Rightarrow$ Include residuals $\hat{u}_{i,t+1}$ as additional regressor in the production function

Evidence Supporting the Exclusion Restrictions

TABLE 3—BALANCE TEST FOR THE INSTRUMENTAL VARIABLES

	log toy price	log food price	Conflict
Mother's cognitive skill	0.021	0.007	−0.004
	(0.009)	(0.007)	(0.003)
Mother's socio-emotional skill	−0.006	−0.014	0.005
	(0.008)	(0.006)	(0.005)
Mother is married	0.013	0.030	0.017
	(0.016)	(0.012)	(0.008)
Wealth index	−0.003	0.002	0.002
	(0.007)	(0.006)	(0.004)
Terrorism	0.015	−0.012	0.005
	(0.014)	(0.013)	(0.009)
Constant	8.025	8.062	0.049
	(0.028)	(0.017)	(0.007)
Observations	1,010	1,023	1,023
R^2	0.011	0.021	0.017
F-statistic	1.656	2.147	1.438
F-test p-value	0.154	0.0671	0.219

Notes: All right-hand-side variables measured at baseline. Asymptotic standard errors in parentheses allowing for clustering at the municipality level. *Terrorism* is the number of terrorist attacks between conception of child and baseline. *Conflict* is the number of conflicts against civil population divided by the municipality population (in thousands) when the mother herself was a child.

Estimation in Three Steps

$\Rightarrow$ Estimate the cov. of the factors based on cov. of the observed measures

$\Rightarrow$ Use these estimates to predict factor scores: $\hat{\theta}^k_{i,t}, \hat{\theta}^k_{i,t}$

$\Rightarrow$ Use predicted scores as data to estimate the parental inv. and prod. function

- Compute conf. intervals for test statistics using the block bootstrap

Investment Functions

Table 5—Estimates of the Material and Time Investment Equations

	Instruments: prices and conflict		Instruments: prices only
	Material investment	Time investment	Material investment
Intercept	−0.015 [−0.114, 0.078]	0.001 [−0.089, 0.089]	−0.013 [−0.11, 0.078]
Treatment	0.209 [0.038, 0.365]	0.318 [0.155, 0.48]	0.204 [0.037, 0.362]
log child's cognitive skill (t)	0.130 [0.016, 0.246]	0.068 [−0.044, 0.18]	0.132 [0.017, 0.25]
log child's socio-emotional skill (t)	−0.028 [−0.133, 0.087]	0.027 [−0.083, 0.145]	−0.030 [−0.131, 0.088]
log mother's cognitive skill	0.748 [0.582, 0.939]	0.349 [0.162, 0.498]	0.750 [0.583, 0.943]
log mother's socio-emotional skill	0.069 [−0.008, 0.139]	0.022 [−0.06, 0.108]	0.068 [−0.008, 0.139]
log number of children	−0.129 [−0.18, −0.077]	−0.128 [−0.186, −0.072]	−0.128 [−0.18, −0.078]
log toy price	−0.096 [−0.168, −0.027]	−0.020 [−0.085, 0.037]	−0.094 [−0.163, −0.026]
log food price	0.091 [0.006, 0.178]	0.042 [−0.026, 0.121]	0.091 [0.006, 0.178]
Maternal childhood exposure to conflict	−0.009	−0.089	

Production Function of Cognitive Skills

TABLE 6—ESTIMATES OF THE PRODUCTION FUNCTION FOR COGNITIVE SKILLS

Instruments:	OLS	IV			
		Prices, conflict	Prices, conflict, treatment	Prices	Prices and treatment
	(1)	(2)	(3)	(4)	(5)
Intercept	−0.018 [−0.094, 0.053]	−0.019 [−0.111, 0.079]	0.007 [−0.089, 0.104]	0.003 [−0.091, 0.072]	−0.009 [−0.083, 0.058]
Treatment	0.083 [−0.025, 0.192]	0.049 [−0.12, 0.391]		−0.028 [−0.186, 0.156]	
log child's cognitive skill (t)	0.675 [0.589, 0.77]	0.648 [0.544, 0.795]	0.638 [0.522, 0.747]	0.626 [0.525, 0.746]	0.631 [0.533, 0.739]
log child's socio-emotional skill (t)	0.001 [−0.091, 0.087]	0.012 [−0.098, 0.143]	0.019 [−0.094, 0.14]	0.015 [−0.084, 0.126]	0.02 [−0.079, 0.127]
log mother's cognitive skill	0.213 [0.089, 0.35]	−0.075 [−0.456, 0.5]	−0.173 [−0.538, 0.201]	−0.102 [−0.495, 0.291]	−0.094 [−0.45, 0.21]
log mother's socio-emotional skill	0.103 [0.031, 0.173]	0.084 [−0.03, 0.163]	0.063 [−0.035, 0.151]	0.06 [−0.019, 0.152]	0.074 [−0.026, 0.155]
log number of children	0.042 [−0.01, 0.092]	0.085 [−0.07, 0.154]	0.084 [0.011, 0.163]	0.089 [0.002, 0.176]	0.086 [0.023, 0.164]
log material investment	0.088 [0.016, 0.157]	0.594 [0.025, 1.179]	0.784 [0.204, 1.383]	0.542 [0.041, 0.996]	0.516 [0.195, 0.946]
log time investment	0.038 [−0.051, 0.129]	−0.171 [−1.198, 0.312]	−0.311 [−0.985, 0.217]		
Goodness-of-fit: Gap in output between treated and control					
Measured in the data	0.115	0.115	0.115	0.115	0.115

Production Function of Socio-emotional Skills

TABLE 7—ESTIMATES OF THE PRODUCTION FUNCTION FOR SOCIO-EMOTIONAL SKILLS

Instruments:	OLS	IV			
		Prices, conflict	Prices, conflict	Prices, conflict, treatment	Prices, conflict, treatment
	(1)	(2)	(3)	(4)	(5)
Intercept	−0.009 [−0.08, 0.063]	−0.02 [−0.087, 0.064]	−0.006 [−0.084, 0.066]	−0.02 [−0.096, 0.058]	−0.022 [−0.089, 0.051]
Treatment	−0.011 [−0.124, 0.093]	−0.071 [−0.318, 0.166]	−0.116 [−0.31, 0.115]		
log child's cognitive skill (t)	0.106 [0.018, 0.192]	0.074 [−0.019, 0.21]	0.09 [−0.017, 0.194]	0.094 [0.003, 0.22]	0.099 [0.002, 0.193]
log child's socio-emotional skill (t)	0.522 [0.403, 0.659]	0.513 [0.374, 0.672]	0.499 [0.389, 0.663]	0.516 [0.387, 0.669]	0.512 [0.403, 0.656]
log mother's cognitive skill	−0.077 [−0.217, 0.049]	−0.146 [−0.443, 0.297]	−0.15 [−0.349, 0.084]	−0.018 [−0.315, 0.328]	−0.083 [−0.231, 0.078]
log mother's socio-emotional skill	0.037 [−0.058, 0.119]	0.035 [−0.062, 0.141]	0.048 [−0.053, 0.135]	0.049 [−0.054, 0.134]	0.043 [−0.042, 0.126]
log of number of children	0.099 [0.047, 0.153]	0.127 [0.017, 0.236]	0.133 [0.034, 0.223]	0.101 [0.028, 0.168]	0.101 [0.036, 0.163]
log material investment	0.154 [0.06, 0.256]	0.015 [−0.621, 0.428]		−0.144 [−0.688, 0.34]	
log time investment	0.109 [−0.006, 0.213]	0.487 [−0.177, 1.258]	0.549 [−0.022, 1.147]	0.448 [−0.125, 1.057]	0.324 [0.025, 0.691]
Goodness-of-fit: Gap in output between treated and control					
Measured in the data	0.087	0.087	0.087	0.087	0.087

Interpreting the Impact of the Intervention

- Cognitive (Socio-emotional) development is mainly determined by
  - Initial cognition (measured at ages 1-2)
  - Stimulation provided by older siblings
  - Material (time) investments provided by parents

$\Rightarrow$ The main channel of the intervention is a shift in parental investments

Complementarities between Inputs

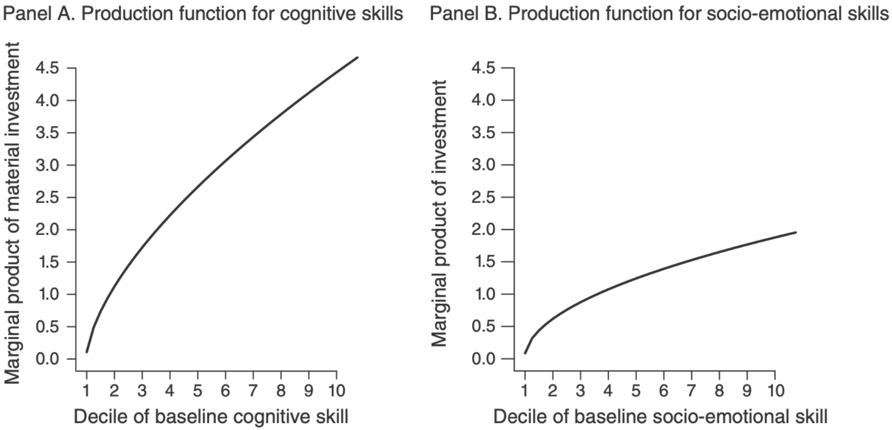


FIGURE 2. COMPLEMENTARITY BETWEEN INVESTMENTS AND BASELINE SKILLS

Implications for longer-term outcomes

- Parental crowding-in suggests treatment will diverge in the medium/long run
- But also possible fade-out due to mean reversion
- Follow-up data (2 years after) suggests partial fade out

$\Rightarrow$ How can we permanently improve parental practices ?

RCT $\Leftrightarrow$ Economic Model

- Model gives a lens to interpret how the intervention affected child development

$\Rightarrow$ Sometimes the experimental variation is what you want to explain

- Need additional sources of variation beyond the experiment